Dynamic Programming Practical Part A

Dynamic programming approaches can be used to solve the task of identifying the shortest path between two elements in a grid. This is similar to the image segmentation example shown in class. This algorithm is often used in Bioinformatics for identifying the shortest path through a graph representing a biological network or signalling pathway or other set of interconnected components. Several algorithms exist to solve this problem, but here we will use dynamic programming to find the shortest path across an array of elements.

3	3	3	3	4	5	6	7	8	9
2	2	2	3	4	5	6	7	8	9
1	1	2	3	4	5	6	7	8	9
From	1	2	3	4	5	6	7	8	9
1	1	2	3	4	5	6	7	8	9
2	2	2	3	4	5	6	7	8	9
3	3	3	3	4	5	6	7	8	9
4	4	4	4	4	5	6	7	8	9
5	5	5	5	5	5	6	7	8	9
6	6	6	6	6	6	6	7	8	To Here

The above example shows the basic concept –although in this case finding the shortest path is not very challenging. Imagine we would like to work out the shortest path between **`From`** and **`To Here`**. To solve this problem we simply work out the number of steps required to go between `From` and `To Here`. We record all possible first step positions, and then from each of those we make another step and record the total number of steps take to reach that square (blue) and this is repeated (green) and so on until we reach To Here. Then we can use a trace back approach to plot the traversal.

4	4	4	4	5	6	7	8	9	10
3	3	3	4	5	6	7			11
2	2	3		5	6	7			
1		3	4		6	7			
From		4	4		7	7			
1		5	5		7	8			
2		4	5	6		8			
3	***3***	4	5	6		8			
4		**4**	**5**	**6**	***7***	**8**	**9**	**10**	**11**
5		5	5	6	7	8			**To Here**

[Other paths are possible but no paths are shorter than the one shown].

- How might you use a similar algorithm to find the shortest path across the above table- avoiding the filled squares. Could you modify the algorithm so that the pathway score takes into account awards encountered along the way.

Just modify the scoring so that when a path encounters an award, the step counter is reduced. In this way the rewards will be accounted for in the path score.

Dynamic Programming Practical Part B

There are some extra considerations when interpreting results from alignment algorithms such as Smith Waterman or Needleman-Wunsch. Here is an example of one additional major consideration.

Installed on the servers are alignment programmes from the EMBOSS sequence analysis suite (http://emboss.sourceforge.net/) that offers many different sequence alignment algorithms. Some useful programmes for us are called `water`, `needle` and `dotmatcher`. These provide tools to align larger and more complex sequences using dynamic programming algorithms. To run these programmes just type their name on the command line, if you type " `-h`" after the name you will be given a help page.

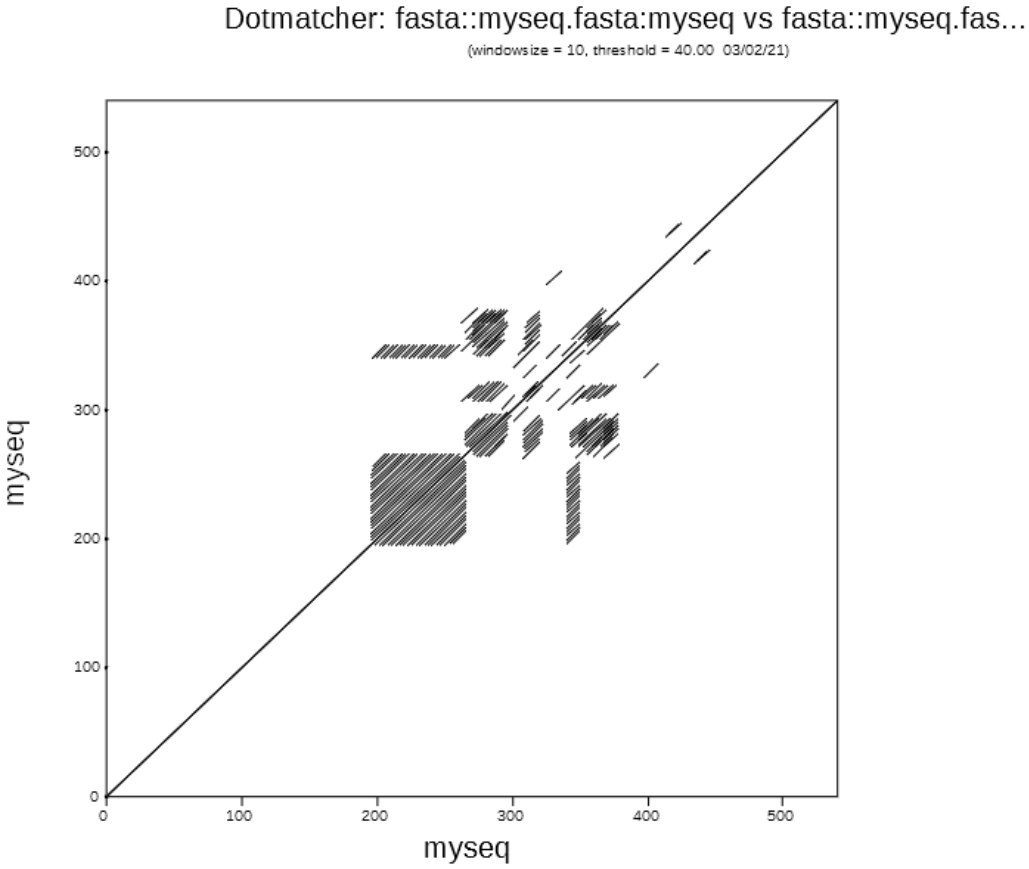
- Run the `water` programme on the two example sequences `myseq.fasta` and `myseq2.fasta`. This will generate a local alignment from these two input files. What are your conclusions from this alignment about the relationship between these sequences?
    - **The two sequences share extensive regions of alignment**
- Now run `dotmatcher` that generates a "dot plot" from the sequences, where each dot in the panel represents one base pair that is identical between sequences. To eliminate noise this dot plot is them smoothed (to remove very short segments with identical sequence) and only runs of identical sequence above a certain threshold score are retained.
- Look at the alignment and the dot plot. What do they tell you about these sequences?
- [Those interested in the biology]. If you want to try to understand the detailed biology of these sequences, the first step is to search DNA sequence databases using Blast. Once you know identity of the sequence, this should give additional meaning to the alignment results.
    - **Even when this sequence is compared to itself the dotmatcher plot shows lines off the diagonal. This is where regions of sequence compare to multiple places in the sequence- so these are repetitive sequences**
    - **These repeat reqions might be identified from the alignment table eg by searching for multiple local alignments with the same sequence.**
    - **This gene is from the HTT Huntinton gene. Change in the number of repeats in this gene results in Huntington's disease. There are many other disorders- called repeat expansion diseases, that similarly have repetitive regions in genes that expand (or delete) and this results in a disorder. (see https://www.ncbi.nlm.nih.gov/pmc/articles/PMC6485936/)**

Example run command for dotmatcher

Note that here I have just compared one sequence (myseq.fasta) to itself!

```
dotmatcher myseq.fasta myseq.fasta -threshold 40 -graph svg
```

If the graph is sent to X11 it send the plot X Window screen. If the graph is set to svg it writes a Scalable Vector Graphics file that can be read by many different image software packages.


Dotmatcher: fasta::myseq.fasta:myseq vs fasta::myseq.fas...
(windowsize = 10, threshold = 40.00 03/02/21)
myseq
myseq
0
100
200
300
400
500